

Thrombotic and thromboembolic events were not associated with tranexamic acid in three large randomized controlled trials

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BACKGROUND:	Thrombotic and thromboembolic events are a common and potentially preventable complication in multitrauma patients, and substantial quality improvement efforts are directed at prevention. The results of several randomized controlled trials (RCTs) related to the association between tranexamic acid (TXA) and thrombotic/thromboembolic events have demonstrated conflicting results. We aimed to address this by examining whether prehospital TXA was associated with higher rates of thrombotic/thromboembolic events in a harmonized data set from three large multicenter RCTs.
METHODS:	We analyzed data using a harmonized data set from three RCTs examining the effects of prehospital TXA: The Pre-Hospital Anti-fibrinolytics for Traumatic Coagulopathy and Hemorrhage Study (PATCH trial), Study of Tranexamic Acid During Air and Ground Medical Prehospital Transport Trial (STAAMP trial) and the Pre-hospital TXA for TBI trial, part of the Resuscitation Outcomes Consortium (ROC trial). Outcomes included deep venous thrombosis (DVT), pulmonary embolism (PE), myocardial infarction, stroke, combined venous thrombotic/thromboembolic events (VTE), and combined arterial thrombotic/thromboembolic events. Multivariable regression was used to adjust for TXA administration, sex, age, injury severity score, Glasgow Coma Scale, shock index, and 24-hour red cell transfusion.
RESULTS:	There were no differences in myocardial infarction, stroke, arterial thrombotic/thromboembolic events, DVT, PE, or VTE in patients who were randomized to TXA compared with those who were not. On univariate analysis, rates of PE, DVT and VTE were significantly higher in the PATCH cohort compared with STAAMP and ROC cohorts, but patients in PATCH had significantly higher injury severity scores and chest trauma when compared with those in ROC and STAAMP.
CONCLUSION:	This multicenter database combining three large RCTs showed that randomization to TXA was not associated with higher rates of arterial and VTE. The higher rates of thrombotic/thromboembolic events observed in the PATCH trial may be explained by higher injury severity as well as protocolized screening. (<i>J Trauma Acute Care Surg.</i> 2026;101: 48–56. Copyright © 2026 Wolters Kluwer Health, Inc. All rights reserved.)
LEVEL OF EVIDENCE:	Sub-analysis of randomized controlled trials; Level II.
KEY WORDS:	Tranexamic acid, deep venous thrombosis, pulmonary embolism, venous thrombotic/thromboembolic events, arterial thrombotic/thromboembolic events

Tranexamic acid (TXA) is an antifibrinolytic agent that is commonly used off-label in trauma with the intent to reduce hemorrhage.¹ As the purpose of TXA is to reduce clot breakdown, the risk of pathologic clotting has remained a concern. Multiple randomized controlled trials (RCTs) including the CRASH-2 trial, The Resuscitations Outcomes Consortium Prehospital TXA for

TBI Trial (ROC),² and Study of Tranexamic Acid During Air and Ground Medical Prehospital Transport Trial (STAAMP)³ did not show significant differences in thrombotic complications with TXA administration, nor did a study based on a harmonized data set from these two trials.⁴ The Pre-Hospital Anti-fibrinolytics for Traumatic Coagulopathy and Haemorrhage Study

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Victoria (R.M.); Department of Surgery, University of Chicago, Chicago, IL (S.R.); Uniformed Services University of the Health Sciences (M.A.S.). This study was presented at the 39th annual meeting of the Eastern Association for the Surgery of Trauma, January 22, 2026, in Atlanta, GA. Supplemental Digital Content is available for this article. Direct URL citations are provided in the HTML and PDF versions of this article on the journal's website, www.jtrauma.com.

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(PATCH)⁵ showed a higher absolute rate of thrombotic complications with randomization to TXA, but this association was not significant.

The retrospective Military Application of Tranexamic Acid in Trauma Emergency Resuscitation trial (MATTERs)⁶ demonstrated a significant association of TXA with thrombotic complications in both the overall study population and patients receiving massive transfusion. The RCT Immunologic Effect of Early Intravenous Two and Four Gram Bolus Dosing of Tranexamic Acid Compared with Placebo in Patients with Severe Traumatic Bleeding trial (TAMPIT)⁷ showed a dose-dependent association of TXA with thrombotic events that approached significance. In addition, an EAST multicenter trial on this subject⁸ showed a significant association of TXA with lower venous thrombotic/thromboembolic events (VTE) rates on unadjusted analysis, although the significance was lost on multivariable adjustment. This wide variability of results is echoed in other observational studies on TXA.^{5,9–13} Arterial thrombotic events (ATE) have been shown to have no association with TXA in some studies^{5,14} but in some studies wherein ATE were grouped with VTE, significant associations with TXA were demonstrated.^{15–17}

The difference in findings between studies is likely multifactorial, with dose, population demographics, and power to detect rare events playing a role. To attempt to address this, we examined thrombotic complications in a harmonized data set including three large RCTs: ROC, STAAMP, and PATCH. We hypothesized that there would not be a significant association of thrombotic complications with randomization to TXA.

METHODS

This study was performed utilizing a harmonized data set from three different large double-blinded, randomized, prehospital TXA trials: The Pre-Hospital Anti-fibrinolytics for Traumatic Coagulopathy and Haemorrhage Study (PATCH trial), Study of Tranexamic Acid During Air and Ground Medical Prehospital Transport Trial (STAAMP trial) and the Prehospital TXA for TBI trial, part of the Resuscitations Outcomes Consortium (ROC trial). Each of the included trial protocols was approved by its respective institutional review boards and was carried out in accordance with relevant guidelines and regulations. Complete protocols are available as supplemental data in the online publications of the trial results. Trial data were harmonized without additional data or laboratory values being collected. Due to the inherent heterogeneity in TXA administration between trials, patients were analyzed primarily on their receipt of TXA vs placebo. Statistical analysis was performed utilizing R (R Core Team). A Consort checklist (Supplemental Digital Content 2, <http://links.lww.com/TA/F316>) and Consort diagram (Fig. 1) were included per EQUATOR guidelines for reporting on RCTs.

Outcomes included the development of thrombotic complications individually with the development of myocardial infarction (MI), stroke, combined ATE, deep venous thrombosis (DVT), pulmonary embolism (PE), combined VTE, and any thrombotic/thromboembolic complication. Patient characteristics including initial emergency department (ED) Glasgow Coma Scale (GCS), initial ED systolic blood pressure, initial ED heart rate, initial ED shock index, age, sex, 24 hour blood transfusion requirements (in units), presence of penetrating

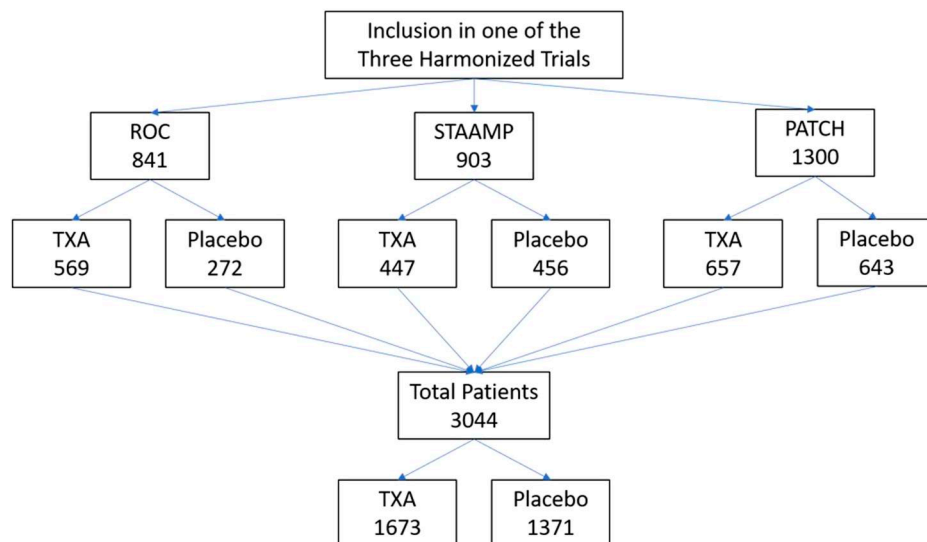


Figure 1. Consort diagram.

TABLE 1. Patient and Injury Characteristics by Receipt of TXA and Study

	ROC			STAAMP			PATCH			Overall	
	Placebo (n = 272)	TXA (n = 569)	Overall (n = 841)	Placebo (n = 456)	TXA (n = 447)	Overall (n = 903)	Placebo (n = 643)	TXA (n = 657)	Overall (n = 1300)	Placebo (n = 1371)	All (n = 3044)
Age in years (median ± IQR)	35 (24–52)	39 (26–56)	37 (25–35)	39 (26–56)	39 (26–53)	39 (26–55)	42 (27–58)	41 (26–61)	41 (27–59)	40 (26–56)	40 (26–57)
Female sex (%)	22.9 (n = 62)	26.3 (n = 149)	25.2 (n = 211)	25.2 (n = 115)	26.8 (n = 120)	26.0 (n = 235)	28.6 (n = 184)	30.1 (n = 198)	29.4 (n = 382)	26.4 (n = 361)	27.9 (n = 467)
Penetrating injury (%)	5.9 (n = 16)	2.5 (n = 14)	3.6 (n = 30)	15.4 (n = 70)	17.4 (n = 78)	16.4 (n = 148)	8.6 (n = 55)	6.7 (n = 44)	7.6 (n = 99)	10.3 (n = 141)	8.1 (n = 136)
Blunt injury (%)	94.1 (n = 256)	97.5 (n = 555)	96.4 (n = 811)	84.6 (n = 386)	82.6 (n = 369)	83.6 (n = 755)	91.4 (n = 588)	93.3 (n = 613)	92.4 (n = 1201)	89.7 (n = 1230)	91.9 (n = 1537)
Shock index > 0.9 (%)	20.4 (n = 55)	15.2 (n = 86)	16.8 (n = 141)	35.3 (n = 153)	34.4 (n = 145)	34.9 (n = 298)	51.2 (n = 302)	51.6 (n = 320)	51.4 (n = 622)	39.4 (n = 1230)	36.6 (n = 1061)
ED SBP (median ± IQR)	133 (119–150)	137 (120–158)	135 (120–155)	130 (105–146)	130 (110–142)	130 (110–144)	110 (88–134)	115 (91–133)	112 (90–134)	122 (100–141)	125 (104–142)
ED HR (median ± IQR)	93 (80–109)	92 (79–108)	92 (79–175)	104 (90–118)	105 (92–117)	105 (91–117)	101 (93–122)	105 (82–125)	104 (82–124)	100 (85–118)	100 (84–118)
GCS initial (median ± IQR)	3 (3–10)	5 (3–11)	4 (3–11)	15 (3–15)	15 (9–15)	15 (7–15)	13 (6–14)	13 (6–15)	13 (6–14)	13 (3–15)	12 (3–15)
AIS head (median ± IQR)	3 (2–4)	3 (2–4)	3 (2–4)	0 (0–2)	0 (0–3)	0 (0–2)	2 (0–3)	2 (0–3)	2 (0–3)	2 (0–3)	2 (0–4)
AIS chest (median ± IQR)	0 (0–3)	0 (0–3)	0 (0–3)	0 (0–3)	0 (0–3)	0 (0–3)	3 (2–4)	3 (2–4)	3 (2–4)	2 (0–3)	2 (0–3)
AIS abdomen (median ± IQR)	0 (0–0)	0 (0–0)	0 (0–0)	0 (0–2)	0 (0–2)	0 (0–2)	2 (0–3)	2 (0–3)	2 (0–3)	0 (0–2)	0 (0–2)
AIS extremity (median ± IQR)	0 (0–2)	0 (0–2)	0 (0–2)	1 (0–2)	1 (0–2)	1 (0–2)	2 (0–3)	3 (2–3)	2 (1–3)	2 (0–3)	2 (0–3)
ISS (median ± IQR)	18 (10–29)	18 (9–29)	18 (9–29)	11 (4–22)	13 (5–22)	12 (5–22)	29 (17–38)	29 (18–41)	29 (17–41)	21 (10–34)	21 (10–34)
Received blood products within 24 h (%)	22.4 (n = 61)	17.8 (n = 101)	19.3 (n = 162)	35.5 (n = 148)	31.5 (n = 141)	32.0 (n = 289)	17.7 (n = 114)	16.3 (n = 107)	17.0 (n = 221)	23.6 (n = 323)	22.1 (n = 672)

ED SBP is the first reported systolic blood pressure in the emergency department. ED HR is the first reported heart rate in the emergency department. GCS initial is Glasgow Coma Scale reported in the emergency department. Percentage of patients who received blood products included any patient who had a reported administered blood product (packed red blood cells, fresh frozen plasma, platelets, and cryoprecipitate volume > 0 mL). Within the PATCH study, there were three burn patients who were listed as blunt due to nonpenetrating injury during harmonization of the data sets.

HR indicates heart rate; IQR, interquartile range; SBP, systolic blood pressure.

TABLE 2. Patient Outcomes by Study and TXA Randomization

	ROC			STAAMP			PATCH			Overall		
	Placebo (n = 272)	TXA (n = 569)	p	Placebo (n = 386)	TXA (n = 369)	p	Placebo (n = 643)	TXA (n = 657)	p	Placebo (n = 1371)	TXA (n = 1673)	p
Myocardial infarct (%)	0.4 (n = 1)	0.9 (n = 5)	0.67	0.2 (n = 1)	0 (n = 0)	1	0.6 (n = 4)	1.2 (n = 8)	0.386	0.4 (n = 6)	0.8 (n = 13)	0.259
Stroke (%)	3.7 (n = 10)	3.2 (n = 18)	0.685	0.7 (n = 3)	0 (n = 0)	0.249	2.2 (n = 14)	2.9 (n = 19)	0.483	2.0 (n = 27)	2.2 (n = 37)	0.704
ATE (MI/stroke) (%)	4.0 (n = 11)	4.0 (n = 23)	1	0.9 (n = 4)	0 (n = 0)	0.124	2.9 (n = 18)	4.2 (n = 18)	0.227	12 (n = 33)	12.7 (n = 50)	0.372
Deep vein thrombosis (%)	3.3 (n = 9)	1.6 (n = 9)	0.127	1.5 (n = 7)	2.7 (n = 12)	0.254	12.9 (n = 81)	15.8 (n = 102)	0.151	7.2 (n = 97)	7.4 (n = 123)	0.833
PE (%)	1.8 (n = 5)	1.4 (n = 8)	0.766	1.5 (n = 7)	2.9 (n = 13)	0.181	7.5 (n = 47)	7.3 (n = 47)	0.915	4.4 (n = 59)	4.1 (n = 68)	0.785
Venous thromboembolism (%)	4.8 (n = 13)	2.8 (n = 16)	0.159	2.9 (n = 13)	5.1 (n = 23)	0.09	17.1 (n = 110)	20.2 (n = 133)	0.155	9.9 (n = 136)	10.3 (n = 172)	0.763
Any thrombotic event (%)	8.1 (n = 22)	6.3 (n = 36)	0.383	3.7 (n = 17)	5.1 (n = 23)	0.334	19.6 (n = 123)	23.5 (n = 152)	0.089	12.0 (n = 162)	12.7 (n = 211)	0.578

injury, presence of blunt injury, regional Abbreviated Injury Scale (AIS) scores, and injury severity scores (ISS). Univariate analysis of outcomes based on patient risk factors was performed using Mann-Whitney *U*, χ^2 , and Fisher exact tests as indicated. Since screening ultrasound was protocolized in the PATCH study and not in the ROC and STAAMP study, both study level and use of screening ultrasound were investigated as possible factors.

Multivariable logistic regression was performed to assess patient and injury-related factors. Any patient or injury factor that was noted to have statistical significance of $p < 0.2$ was included in the preliminary model. Aliased factors were identified and removed from the final model. GCS was separated into three categories: 13 to 15, 9 to 12 and < 9 . ISS was treated as a continuous variable consistent with other studies (TRISS score). Due to the high degree of aliasing with abbreviated injury scores within ISS (as would be expected intuitively), regional injury scores were largely not incorporated into the final model for analysis, but regional injury was individually assessed.

RESULTS

Descriptive patient characteristics were notable for a higher penetrating injury rate in STAAMP, lower GCS in ROC, and collectively more severe injuries in PATCH (Table 1). For the overall data set and each individual study, no outcomes examined were significantly different between patients randomized to placebo or TXA (Table 2). Notably, the incidence of any complication in PATCH was 19.6% and 23.5% for the placebo and TXA groups, respectively, compared with 8.1 and 6.3% in ROC and 3.7% and 5.1% in STAAMP. There were no differences in MI, stroke, ATE, DVT, PE, or VTE in patients

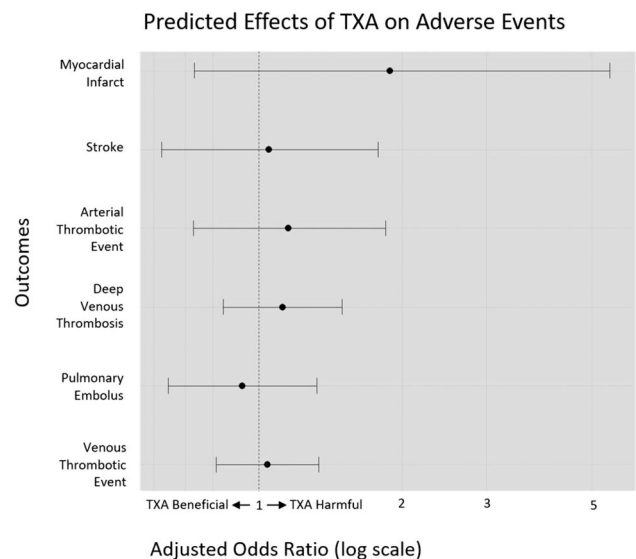


Figure 2. Predicted effects of TXA on adverse effects.

TABLE 3. Univariate Analysis of Outcomes by Patient/Injury

Univariate Analysis of Outcomes by Patient/Injury	MI			Stroke			ATE (MI/CVA)			Deep Vein Thrombosis
	No (n = 2999)	Yes (n = 19)	p	No (n = 1328)	Yes (n = 27)	p	No (n = 2934)	Yes (n = 83)	p	No (n = 2798)
Randomized to receive TXA (%)			0.241			0.658			0.372	
Yes	99.20	0.80		97.80	2.20		97.00	3.00		92.60%
No	99.60	0.40		98.00	2.00		97.60	2.40		92.80%
Screening US protocolized study (%)			0.1			0.126			0.032	
No	96.60	0.40		98.20	1.80		97.80	2.20		97.90
Yes	99.10	0.90		97.40	2.60		96.50	3.50		85.60
Study (%)			0.051			< 0.001			< 0.001	
PATCH	99.10	0.90		97.40	2.60		96.50	3.50		85.60
ROC	99.30	0.70		96.70	3.30		96.00	4.00		97.90
STAAMP	99.90	0.10		99.70	0.30		99.60	0.40		97.90
Age	39 (26–56)	69 (55–76)	< 0.001	39 (26–56)	45 (29–62)	0.094	39 (26–56)	52 (32–69)	< 0.001	39 (26–56)
Sex (%)			0.664			0.06			0.079	
Female	99.30	0.70		97.10	2.90		96.30	3.70		94.10
Male	99.40	0.60		98.20	1.80		97.60	2.40		92.00
Had blunt injury (%)			0.409			0.003			0.001	
Yes	99.30	0.70		97.70	2.30		97.00	3.00		92.40
No	100	0		100	0		100.00	0.00		95.60
SBP initial	125 (104–142)	116 (87–128)	0.131	125 (104–142)	126 (92–138)	0.207	125 (104–142)	122 (91–138)	0.066	126 (105–143)
HR initial	100 (84–118)	94 (76–119)	0.363	100 (84–118)	99 (82–120)	0.753	100 (84–118)	98 (80–121)	0.870	100 (83–117)
GCS	13 (5–15)	13 (3–15)	0.714	5 (3–11)	13 (3–15)	< 0.001	13 (3–15)	6 (3–13)	< 0.001	13 (3–15)
Elevated Shock Index (%)			0.615			0.084			0.083	
> 0.9	99.20	0.80		97.20	2.80		96.40	3.60		88.00
≤ 0.9	99.40	0.60		98.20	1.80		97.60	2.40		95.10
AIS head	2 (0–3)	2 (0–4)	0.952	2 (0–3)	4 (3–5)	< 0.001	2 (0–3)	4 (2–5)	< 0.001	2 (0–3)
AIS chest	2 (0–3)	3 (3–4)	0.016	2 (0–3)	3 (0–4)	0.003	2 (0–3)	3 (0–4)	< 0.001	2 (0–3)
ISS	22 (17–32)	21 (10–34)	0.294	21 (10–33)	34 (26–44)	< 0.001	21 (10–33)	30 (18–41)	< 0.001	19 (9–33)
Blood 24PRBC volume in mL	0 (0–0)	0 (0–0)	0.975	0 (0–0)	0 (0–300)	0.046	0 (0–0)	0 (0–300)	0.081	0 (0–0)
Blood 24PRBC unit	0 (0–0)	0 (0–0)	0.96	0 (0–0)	0 (0–1)	0.046	0 (0–0)	0 (0–1)	0.082	0 (0–0)

HR indicates heart rate; SBP, systolic blood pressure.

who were randomized to TXA compared with those who were not on both univariate (Table 3) and multivariable analyses (Table 4), with regression analysis (Fig. 2) demonstrating the following adjusted odds associated with randomization to TXA: Any VTE 1.04 (95% CI, 0.81–1.34), DVT 1.12 (95% CI, 0.84–1.50), and PE 0.92 (95% CI, 0.65–1.32). On univariate analysis, rates of PE, DVT, and VTE were significantly higher in the PATCH cohort compared with STAAMP and ROC cohorts, but patients in PATCH had significantly higher ISS and AIS Chest when compared with those in ROC and STAAMP. On multivariable analysis, randomization to TXA was not associated with higher odds of venous or ATE (Table 4).

DISCUSSION

In this study, we found no significant association of thrombotic complications with TXA in a harmonized data set from three large randomized trials. While these

results provide reassurance that TXA is a relatively safe therapy, it remains important to contextualize the findings in the setting of conflicting prior findings. In addition, applying these findings to clinical use that does not exactly align with how TXA was administered in these RCTs is somewhat concerning without considering how thrombosis is measured and reported, as well as some important physiologic factors that may augment TXA's effect.

The methodology for measuring and reporting thrombotic events between studies is variable. Distinctions between MI, Stroke, ATE, DVT, PE, and VTE are reported variably as individual events and composite outcomes, which can complicate comparisons between studies. Factors influencing clinical significance including site of DVT (proximal vs distal) and severity of PE (asymptomatic, symptomatic but without hemodynamic compromise, or causing hemodynamic compromise or death) are rarely parsed out in a meaningful way. Most studies, including the three RCTs in this analysis, as well as CRASH-2, collected data on arterial thrombosis including

TABLE 3. Univariate Analysis of Outcomes by Patient/Injury

Deep Vein Thrombosis		Pulmonary Embolism			VTE			Any Thrombotic Event		
Yes (n = 220)	<i>p</i>	No (n = 2891)	Yes (n = 127)	<i>p</i>	No (n = 2735)	Yes (n = 308)	<i>p</i>	No (n = 2643)	Yes (n = 373)	<i>p</i>
	0.794			0.724			0.747			0.578
7.40%		95.90	4.10		89.70	10.30		87.30	12.70	
7.20%		95.60	4.40		90.10	9.90		88.00	12.00	
	< 0.001			< 0.001			< 0.001			< 0.001
2.10		98.10	1.90		96.60	3.70		94.40	5.60	
14.40		92.60	7.40		81.30	18.70		78.40	21.60	
	< 0.001			< 0.001			< 0.001			< 0.001
14.40		92.60	7.40		81.30	18.70		78.40	21.60	
2.10		98.50	1.50		96.60	3.40		93.10	6.90	
2.10		97.80	2.20		96.00	4.00		95.60	4.40	
45 (29–61)	0.003	45 (31–61)	39 (26–56)	0.007	39 (26–56)	45 (30–61)	< 0.001			0.178
	0.064			0.475			0.426	38 (26–55)	45 (25–56)	1
5.90		95.40	4.60		90.60	9.40		87.70	12.30	
7.80		95.90	4.10		89.60	10.40		87.60	12.40	
	0.05			0.038			0.012			< 0.001
7.60		95.60	4.40		89.40	10.60		87.00	13.00	
4.40		98.20	1.80		94.20	5.80		87.60	12.40	
110 (89–135)	< 0.001	126 (105–260)	113 (93–132)	< 0.001	127 (106–143)	111 (91–135)	< 0.001	132 (116–150)	126 (102–146)	0.016528
115 (95–130)	< 0.001	100 (83–118)	112 (92–128)	< 0.001	100 (83–117)	113 (93–129)	< 0.001	99 (84–113)	100 (85–121)	0.225401
11 (5–14)	0.103	13 (3–15)	13 (7–15)	0.16	13 (3–15)	12 (5–14)	0.595	12 (3–15)	3 (3–12)	< 0.001
	< 0.001			< 0.001			< 0.001			< 0.001
12.00		92.80	7.20		83.10	16.90		80.70	19.30	
4.90		97.30	2.70		93.20	6.80		91.10	8.90	
3 (0–4)	0.001	2 (0–4)	2 (0–3)	0.3	2 (0–4)	2 (0–4)	0.05	2 (0–3)	3 (0–5)	< 0.001
3 (2–4)	< 0.001	2 (0–3)	3 (2–4)	< 0.001	2 (0–3)	3 (2–4)	< 0.001	2 (0–3)	3 (2–4)	< 0.001
33 (22–43)	< 0.001	21 (10–33)	27 (19–38)	< 0.001	19 (9–33)	29 (22–41)	< 0.001	14 (5–25)	25 (14–34)	< 0.001
0 (0–300)	0.001	0 (0–0)	0 (0–0)	0.313	0 (0–0)	0 (0–300)	< 0.001	0 (0–0)	0 (0–700)	< 0.001
0 (0–1)	0.001	0 (0–0)	0 (0–0)	0.305	0 (0–0)	0 (0–1)	< 0.001	0 (0–0)	0 (0–2)	< 0.001

MI and stroke. CRASH-2 reported a decrease in fatal “vascular occlusive events,” but this appears to have been driven by reduced incidence of MI in the TXA group.¹⁵ The study showing the strongest association of TXA with thrombosis, TAMPITI, included multiple arterial thrombosis types in their composite outcome, including MI, CVA and mesenteric thromboses.^{7,17} This may have contributed to the higher rate of thrombotic events; however, the relationship between arterial thrombosis and TXA is less well established. Although limited by the inclusion of several retrospective studies, a large meta-analysis of nonsurgical patients found no significant difference in arterial or venous complications with the use of TXA.¹⁴ Conversely, the

HALT-IT RCT which examined TXA use in gastrointestinal bleeding demonstrated that VTE rate was twice as high in the TXA group compared with the placebo group (0.8% vs. 0.4%, statistically significant) but no difference in ATE was observed.¹⁶ In addition to the commonly reported MI and CVA, PATCH also reported on “other arterial events.”⁵ This category did not show absolute or significant differences between TXA and placebo groups in the original study, but when all arterial events were examined as a composite outcome in this harmonized set, they were significantly more common in the PATCH group. It is difficult to assess how much reporting bias affected this result.

TABLE 4. Multivariable Regression for Adverse Events by TXA

	Adjusted Odds Ratio (95% CI)	p
Myocardial infarct	1.885 (0.731–5.453)	0.207
Stroke	1.048 (0.624–1.781)	0.859
ATE (MI/stroke)	1.152 (0.729–1.847)	0.547
Deep vein thrombus	1.121 (0.843–1.496)	0.435
PE	0.922 (0.645–1.322)	0.657
Venous thromboembolism	1.042 (0.814–1.335)	0.747

Although diagnostic modalities for different types of thrombosis vary, DVT is almost universally diagnosed with duplex ultrasound. Screening practices vary widely and whether screening is indicated or what the optimal approach is, remain matters of debate.¹⁸ The proposed benefit of routine screening is to identify and treat asymptomatic DVT before it progresses to symptomatic or fatal PE,¹⁹ but searching for DVTs also leads to finding more DVTs (surveillance bias) and potentially unnecessary cost and risk to patients from anticoagulation.²⁰ In initial analyses of this harmonized data set, we found that DVT and PE rates were significantly higher in PATCH as compared with ROC and STAAMP. On closer examination, we found two important confounders: PATCH patients had a higher average chest AIS, and PATCH was the only study of the three that specified duplex ultrasound screening (between 5 and 7 d post injury) in the protocol. While some sites within ROC and STAAMP perform duplex screening routinely, it was not reported which sites screened and what screening protocols were used. As such, we could not assess screening versus not screening directly outside study effects. The TAMPITI trial which reported a dose-dependent increase in thrombotic events in patients who randomized to TXA used active screening.¹⁷ However, if screening was the only factor influencing DVT rates and if the purpose of screening held true, PATCH should have had a lower incidence of PE. This was not the case. The association of increased incidence of PE with increased injury severity, particularly in the chest, may have led directly to lung injury and primary pulmonary thrombosis, which could not be distinguished from emboli.

In addition to the association of PE (vs. primary pulmonary thrombosis) with severity of chest injury, we found that ISS was significantly associated with all outcomes examined with the exception of MI. Due to coagulation abnormalities and immobility, trauma and trauma severity are well-established risk factors for thrombosis.^{21–23} The majority of studies that have shown a significant association of TXA with thrombotic events have been observational, including MATTERS, and may have incompletely controlled for severity of injuries. A recent observational study including 17 Level 1 centers did show a higher rate of VTE in patients who received TXA, but when corrected for severity of injury no association was seen.¹⁰ Injury severity should be easier to account for with RCTs, but as exemplified with PATCH, that is not always the case. The effect of injury severity and specifically AIS chest was difficult to identify in a patient population that had a relatively high average injury severity, but was clarified with comparison to other RCTs with different patient demographics. Injury type and severity have also been shown to be associated with the patient's fibrinolytic phenotype. In one pediatric study, TXA was shown to be associated with an increased rate of DVT in patients in fibrinolytic shutdown.²⁴ In addition, depending on injury type, particularly comparing TBI to hemorrhagic shock, worse injuries may be associated with different predominant fibrinolytic

phenotypes.^{25–30} It is also important to note that TXA has a stronger association with mortality benefit in more severely injured or hypotensive patients.^{3,6,31} If, as suggested by our results, TXA itself is not driving higher VTE rates in severely injured patients, the benefits of TXA likely outweigh the risks. This may not be the case for less severely injured patients.^{32,33}

Limitations

This study has several limitations. While the data were derived from RCTs, it is a secondary analysis and inherently limited by differences between studies in patient demographics, dosing regimens, VTE screening, and other difficult-to-measure variables. Differences in prehospital dosing regimens in combination with demographics precluded certain analyses, most importantly, the question of a dose-dependent relationship of TXA with thrombotic complications. While a prior study that harmonized ROC and STAAMP was able to compare outcomes between prehospital doses,⁴ the fact that ROC was the only study that included a prehospital 2 g bolus combined with stark demographic differences, particularly between ROC and PATCH raised concerns that the comparison would not be valid. The inability to compare dosages is important as dose may affect both fibrinolytic and nonfibrinolytic effects of TXA that are important to patient outcomes,^{24–27,34} and these interactions may vary between subgroups of patients.³⁵ Although the total number of patients included was high, some subgroups may still be underpowered to show a difference in thrombotic risk. Finally, detailed fibrinolytic data were not available for all studies, so differences in thrombotic risk and other outcomes between fibrinolytic phenotypes could not be examined.

CONCLUSIONS

In this analysis of harmonized data from three large RCTs examining TXA, randomization to TXA was not associated with increase in risk for thrombotic events, indicating it is safe to use within the parameters used in the three RCTs examined in this study. TXA's downstream effects are complex and influenced by multiple factors, and the higher rates of thrombotic/thromboembolic events observed in the PATCH trial may be explained by higher injury severity as well as protocolized screening. Thrombotic/thromboembolic risk for different TXA regimens and smaller subgroups should continue to be examined in future studies.

AUTHORSHIP

A.B., J.K., M.S., J.S., and S.R. participated in conception and study design. A.B. and J.K. participated in literature review, data analysis and interpretation and drafting of the manuscript. M.S., J.S., S.R., R.M., S.B., C.B., R.G., M.N., and B.M. participated in data acquisition and critical revision.

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DISCLOSURE

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